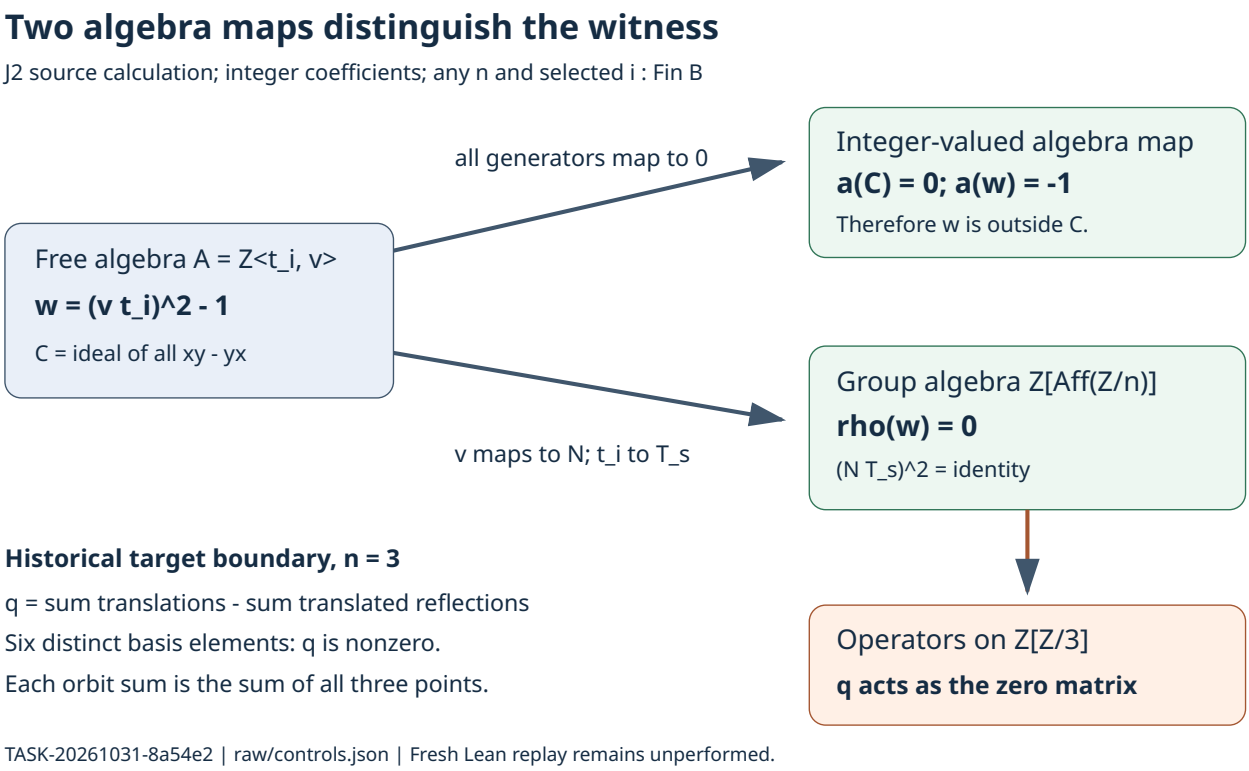
J2: squared negation-translation witness

`TASK-20261031-8a54e2`; joint `J2_commutator_witness`.

Producer PR #2417: `f4330e1a29feb8fdea9f1eab7131ae5cde2e2b6f`.
Plan PR #2847: `774a6a86f181740285ed9834a648b40fb9aa8ece`.

The companion witness has augmentation -1 and affine group-algebra image 0, including $B = 1, n = 2$. Its source-declared statements faithfully express that result. Exact finite controls passed. Fresh Lean replay was not allocated to this reviewer, so the formal-review outcome remains **inconclusive**; the source-fidelity finding is **faithful scoped**.



Exact declarations. In namespace `CryptoResearch.NCP`:

```
neg_translate_square_not_in_commutators
neg_translate_square_in_action_kernel
```

These declarations concern the same element $(\nu t_i)^2 - 1$. The first is universally quantified in B and $i : \operatorname{Fin} B$; the second also in $n : \mathbb{N}$ and $s : \operatorname{Fin} B \to \operatorname{ZMod} n$. The target is $\mathbb{Z}[\operatorname{affineGroup}(\operatorname{ZMod} n)]$, with the affine group a subgroup of actual permutations. No oddness, primality or surjectivity assumption occurs.

Calculation. The algebra map a sending every free generator to zero kills every $xy - yx$ and hence the two-sided ideal C they generate. But $a((\nu t_i)^2 - 1) = -1$. Under permutation composition, $NT_s(x) = -(x + s)$ and $(NT_s)^2(x) = x$. Extending to group-algebra basis elements therefore sends the witness to zero. At $n = 2, s = 1$, N is identity, T_s is the transposition, and its square is identity. At $B = 0$ there is no selected i; the older $\nu^2 - 1$ witness still applies.

Controls and the historical target boundary

Exact controls. The standalone integer/permutation checker imports no producer code. It checks all translations for moduli $1, \ldots, 5$: 15 cases and 55 point evaluations. All squared witnesses map to zero and have zero-generator augmentation -1. Removing the square fails at $n = 2, s = 1$. Sending generators to 1 also fails to separate the witness. At $n = 3, s = 1$, $\nu t - t\nu$ has a nonzero group-algebra image: the witness establishes $K \neq C$, not $C \subsetneq K$.

The $n = 2$ full-presentation control. For the historical ideal J, the substitution $t \mapsto 1, \nu \mapsto -1$ kills translation commutators, $\nu^2 - 1$, $(\nu t)^2 - 1$, and all lattice binomials. It maps $\nu - 1$ to -2. But $\nu - 1$ is in the action evaluation kernel when $n = 2$. Thus the full historical equality does not extend to $n = 2$. The exact producer note already states this boundary; the companion witness remains valid there.

A separate historical bridge breaks at odd order. For $G = \mathbb{Z}/3$, set

$$q = (\delta_1 + \delta_T + \delta_{T^2}) - (\delta_N + \delta_{TN} + \delta_{T^2N}).$$

Its six basis elements are distinct, so $q \neq 0$ in the group algebra. Both sums send every basis point to the sum of all three points, so q acts as zero on $\mathbb{Z}[G]$. The raw control records all six coefficients and the 3×3 zero matrix. A nonsingular realization is $E/\mathbb{F}_5 : y^2 = x^3 + 4x + 2$, with affine points $(3, 1), (3, 4)$ and infinity, so $E(\mathbb{F}_5) = G$ has order 3. The control checks point enumeration and nonsingularity, not the elliptic-curve group law. Thus the historical note's group-algebra inclusion into point operators is false; a faithful permutation group action need not give an injective algebra representation.

Additional obligations for full kernel equalities. Define the free group and historical ideal J; prove the dihedral normal form and exact lattice kernel; prove all of J maps to zero; prove the converse by a quotient basis or a justified finite-rank argument; supply sign faithfulness and generation hypotheses; bridge the cyclic model to the curve. A claim about point operators must instead use the actual representation map and account for its additional kernel.

Verification limits and next action. No Lean/Lake process, axiom audit, kernel rechecker, experiment or cloud job was launched. Parent dispatch retained heavy-build ownership elsewhere. Memory was not measured to be scarce. Exact-pin named-declaration replay, elaborated types, complete input manifests and transitive axiom checks remain to be supplied by an owned verifier. The historical embedding sentence requires a new producer revision separating the two target algebras. This packet changes no official scientific status.

Sources and reproducibility. See `../review.md` for the full source-linked report; `source_manifest.json` binds every source by SHA-256 and revision. Producer sources are `CommutatorCompanion.lean`, `CommutatorIdeal.lean` and `THM_COMMUTATOR_KERNEL1.md`, retained under `inputs/`. The pinned mathlib primary sources for FreeAlgebra, TwoSidedIdeal.Kernel and permutation composition were inspected at `cfc55fc58714a59dc06e6e79edc6d1d95140d3c2`. Toolchain: Lean 4.35.0. Independent arithmetic output: `controls.json`; command and hashes: `controls_receipt.json`. No sibling report was read. Sole archive owner: `TASK-20261031-20ed91`.